William Te Ata O Tū MacDonald

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Work Experience

Shift Manager, Supa Karts 2022 – present | Christchurch
- Promoted from track staff to Shift Manager after demonstrating leadership and comprehensive operational knowledge.
- Responsible for safe track operation, staff coordination and resolving customer and operational issues during busy shifts.

Lifeguard, Waimakariri District Council 10/2024 – 02/2025 and 11/25–2/26
- Maintained safe aquatic operations while providing customer service and responding to incidents.

Projects

Elevator system control
- Developed ladder logic and structured text in Omron CX Programmer to control a scaled down elevator using encoder feedback and PI position control. Scored 93%.

Line following robot
- Designed a PCB in KiCad that passed DRC and ERC checks, alongside a 15:1 spur gear transmission in SolidWorks for a line-following robot under material and dimensional constraints. Testing with 3D printed parts revealed excess inertia, so the design was switched to a friction drive instead.
- Produced a technical design report documenting design decisions and achieved 90%.

CAD/CAM coffee coaster
- Designed and generated G-code toolpaths to machine a coffee coaster from recycled 3D-printing waste.

Education

Bachelor of Engineering with Honours 2024 – present
Mechatronics engineering, University of Canterbury

Certificates

- Current first aid qualification

Technical skills

Manufacturing
Manual lathe, milling, fabrication.

CAD/CAM
SolidWorks, G-code, KiCad

Programming
C (embedded systems, AVR), Python (numpy, matplotlib)

Controls
Omron PLC, ladder logic, structured text

References

Available upon request